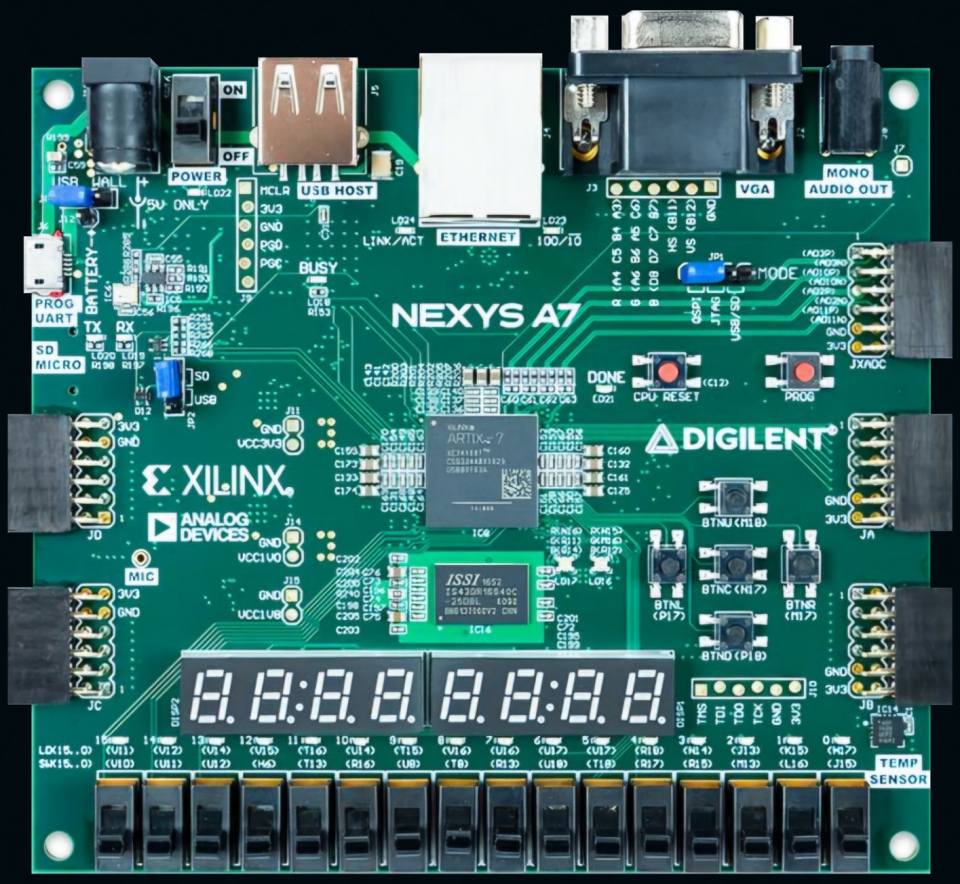
LED PING-PONG

FPGA-BASED GAME ON NEXYS A7

PROJECT DESCRIPTION

This project is part of the BPC-DE1 course at the Faculty of Electrical Engineering and Communication Technologies. It implements a simple LED Ping-Pong game on an FPGA board. Players control left and right paddles using buttons, and the game keeps track of scores. The goal is to reach the defined winning score first.



TEAM MEMBERS

Frank Patrik

Programming & Overall program structure

Hromek Matěj

Timing module & Higher difficulty settings

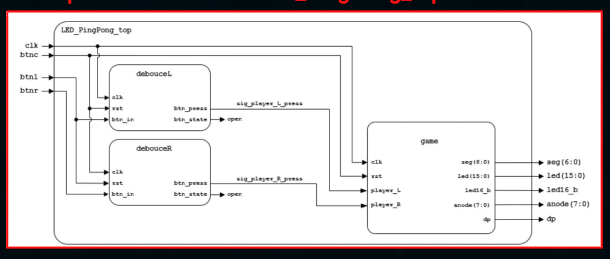
Križan Damián

Implementation & Documentation

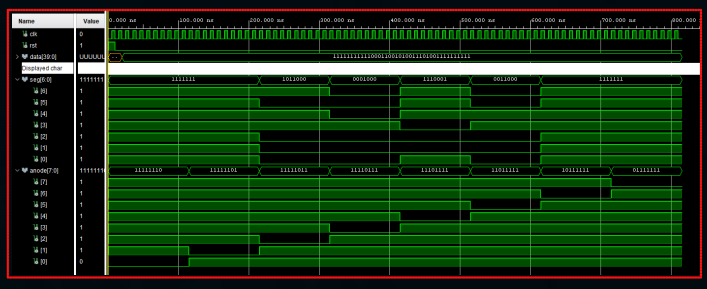
Toman Jan

Score display on 7-segment

The top-level VHDL file is LED_PingPong_top.vhd:

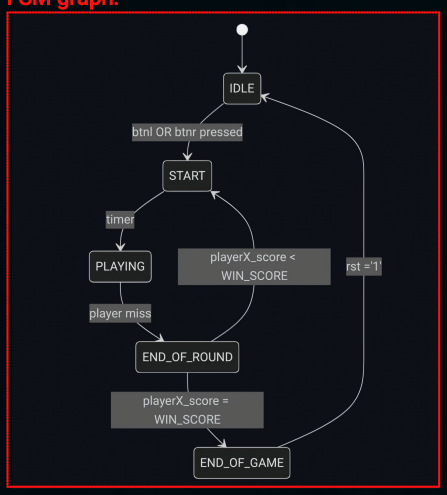
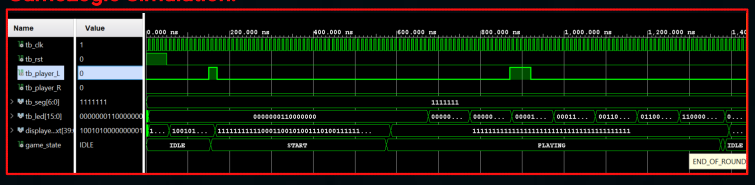


7-Segment Display Multiplexing:



Binary to 7-Segment Decoder (bin2seg):

FSM graph:



GameLogic Simulation: